

Mini Project Report

Discipline of Computer Science and Technology
Goa Business School, Goa University



Smart Fish Disease Detection & Identification System

Submitted in partial fulfillment of the requirements for the course Fundamentals of Artificial Intelligence CSA 5013

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ABSTRACT

The aquaculture industry, spanning from commercial pisciculture to retail aquaria, relies heavily on fish health as a primary driver of economic sustainability. However, the scarcity and high cost of professional aquatic pathologists often leave small-to-midscale operators without specialized diagnostic support, leading to frequent misdiagnosis and delayed treatment. This project proposes a **Smart Fish Disease Detection & Identification System** designed to bridge this gap. By leveraging advanced **Convolutional Neural Networks (CNNs)**, the system automates the extraction of high-level spatial features—such as lesions, textures, and scales—directly from fish imagery. We conduct a comparative analysis of multiple CNN architectures to identify the optimal balance between diagnostic accuracy and computational efficiency. The framework is trained to classify diseases into four major categories: **Bacterial, Viral, Fungal, and Parasitic**. Performance is evaluated using convergence curves and F1-scores to ensure robustness across varying water conditions and image qualities. Finally, the model is deployed via a web-based framework, providing accessible, real-time diagnostic reports. This system empowers non-expert caregivers to intervene early, reducing economic loss and enhancing the sustainability of freshwater aquatic life.

INTRODUCTION

In recent years, the aquaculture sector has become a vital component of the global food economy and the ornamental fish trade. As the industry scales, the importance of **Health Diagnosis** has surged, driven by the need to maintain operational efficiency and safety in delicate aquatic environments. Even minor environmental fluctuations can trigger rapid disease outbreaks, leading to catastrophic biological and financial losses.

Despite these risks, the industry faces a significant bottleneck: the lack of accessible diagnostic expertise. Professional aquatic veterinarians are often prohibitively expensive or geographically unavailable for small-scale farmers and retail store owners. Consequently, these caregivers rely on subjective visual inspections. Because different pathogens often manifest in overlapping physical symptoms, manual diagnosis is prone to human error, and early-stage infections are frequently overlooked.

Traditional diagnostic methods, while accurate in laboratory settings, are often too time-consuming to address the rapid progression of aquatic diseases. To overcome these limitations, there is a pressing need for intelligent, automated tools. This project introduces a Deep Learning-based approach that utilizes CNNs to perform **Automated Feature Learning**. Unlike traditional handcrafted methods that struggle with the nonlinear complexities of biological data, CNNs can autonomously learn spatial hierarchies and patterns from imagery. By integrating these models into a functional web application, this research aims to democratize expert-level diagnostics, allowing for rapid intervention and more resilient aquaculture management.

PROBLEM STATEMENT

In the aquaculture industry, with environments ranging from commercial pisciculture ponds to retail fish store aquariums, fish health is the primary determinant of economic success. Professional aquatic veterinarians and pathologists are scarce and prohibitively expensive for small-to-midscale farmers and aquariums. This leaves them without specialized diagnostic support. Non-expert caregivers often rely on subjective visual inspections. Due to the overlapping physical symptoms of different pathogens, misdiagnosis is frequent or early symptoms could be overlooked. Traditional diagnostic methods are time-consuming and delay diagnosis. The goal of this project is to bridge the gap between expert knowledge and daily fish care by developing a Smart Fish Disease Detection & Identification System. By leveraging advanced Convolutional Neural Network (CNN) architectures, this system will automate the feature learning process to provide rapid, reliable, and accessible diagnostic classifications. This will empower farmers and store owners to intervene early and ensure the sustainability of freshwater aquatic life without the immediate need for expert care.

Objectives:

- **Automated Feature Learning:** Develop a Deep Learning framework to learn high-level representations and spatial hierarchies (edges, textures, lesions) directly from fish disease imagery.
- **Comparative Architecture Analysis:** Conduct a comparative study of multiple CNN architectures to determine the optimal balance between diagnostic accuracy and computational efficiency.
- **Multi-Category Classification:** Train and validate the system to identify major disease categories, including Bacterial, Viral, Fungal, Parasitic.
- **Performance Evaluation:** Analyze model reliability through convergence curves and metrics such as Accuracy and F1-score to ensure generalizability across different water conditions and imaging qualities.
- **Deployment via Web Framework:** Build and deploy a functional web application to serve the model, allowing users to obtain automated diagnostic reports from uploaded images.

EXISTING SYSTEM

1. Traditional and Manual Assessment Methods (Standard Practice)

The majority of small-to-midscale farmers and hobbyists continue to rely on these methods, despite their significant susceptibility to error.

- **Visual Observation:** Caregivers identify "clinical signs," such as the presence of white spots (Ichthyophthiriasis), frayed or eroded fins (Fin Rot), or "flashing" (the behavior of fish scraping against substrata).
- **The Limitation:** This approach is **highly subjective**. Numerous diseases present with visually identical symptoms. For instance, a fungal infection and a bacterial ulcer may both manifest as white dermal patches, yet they necessitate entirely distinct pharmacological treatments.
- **Reference Materials:** Utilization of printed texts or static online resources for self-diagnosis.
- **The Limitation:** This process is dependent on the user's capacity to accurately characterize the observed pathology. An inaccurate identification of a symptom will invariably lead to an incorrect diagnostic conclusion.

2. Clinical and Laboratory Diagnostics (The Professional Standard)

These methods represent the "gold standard" for diagnostic accuracy but are frequently inaccessible to the general populace.

- **Microscopy and Biopsy:** Involves obtaining a superficial skin scraping or a gill clipping for microscopic examination to identify ectoparasites.
- **PCR and Histopathology:** Requires sending tissue samples to a specialized laboratory for the molecular identification of viral or bacterial DNA.
- **The Limitation:** These procedures are often **invasive** (occasionally necessitating euthanasia of the specimen), **cost-prohibitive**, and **time-intensive**. In regions such as Goa, locating a specialized aquatic veterinarian and enduring a 3–5 day period for laboratory results frequently results in the systemic infection of the entire aquaculture unit by the time the diagnosis is rendered.

3. Rule-Based Digital Systems (Legacy Applications)

Existing applications (e.g., [FishVet](#)) function as digital "decision trees".

- **Operational Mechanism:** Users select symptoms from a predefined checklist (e.g. Gill hyperemia, Anorexia), and the application subsequently computes the most probable etiology.

The Limitation: These systems are fundamentally **inflexible**. They do not perform image recognition of the fish; they merely process the data input by the user. Consequently, they are incapable of detecting the subtle nuances in tissue texture or lesion morphology that a Convolutional Neural Network (CNN) can discern.

PROPOSED SYSTEM

The proposed system consists of an advanced CNN-driven diagnostic framework designed to identify freshwater fish diseases by leveraging a multi-architecture approach that includes ResNet-50, MobileNetV2, DenseNet-121, and EfficientNet-B0. By utilizing a FastAPI backend, the system loads all four specialized Convolutional Neural Network architectures into memory during server initialization, allowing for rapid, asynchronous inference through a unified PyTorch execution environment. Each uploaded image undergoes a standardized pre-processing pipeline involving resizing to 224×224 pixels and normalization using ImageNet statistics to ensure consistent feature extraction across the models.

This computational depth is presented through a professional, minimalist frontend where the user interacts with a grid layout that displays the comparative predictions and confidence percentages of each model side-by-side, providing a transparent view of the AI's decision-making process. This dual-layered architecture provides a robust fail-safe mechanism where discrepancies between models serve as a diagnostic indicator for potential secondary infections or complex cases, ultimately offering a more reliable and transparent decision-making tool for aquatic health management.

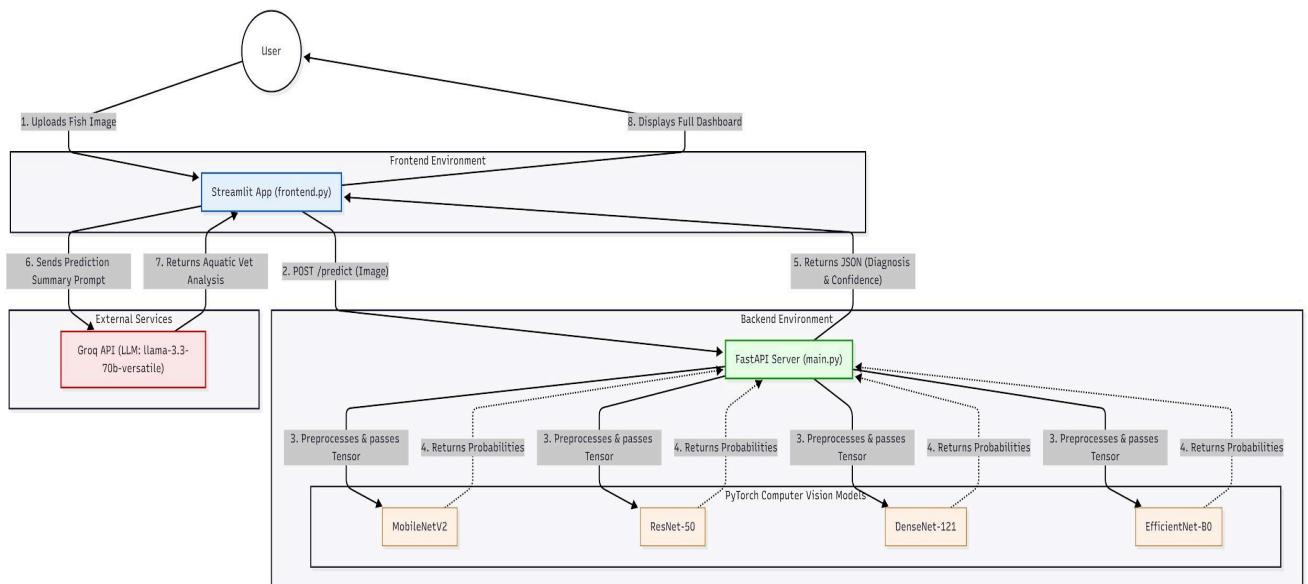


Figure 1.1: System Architecture

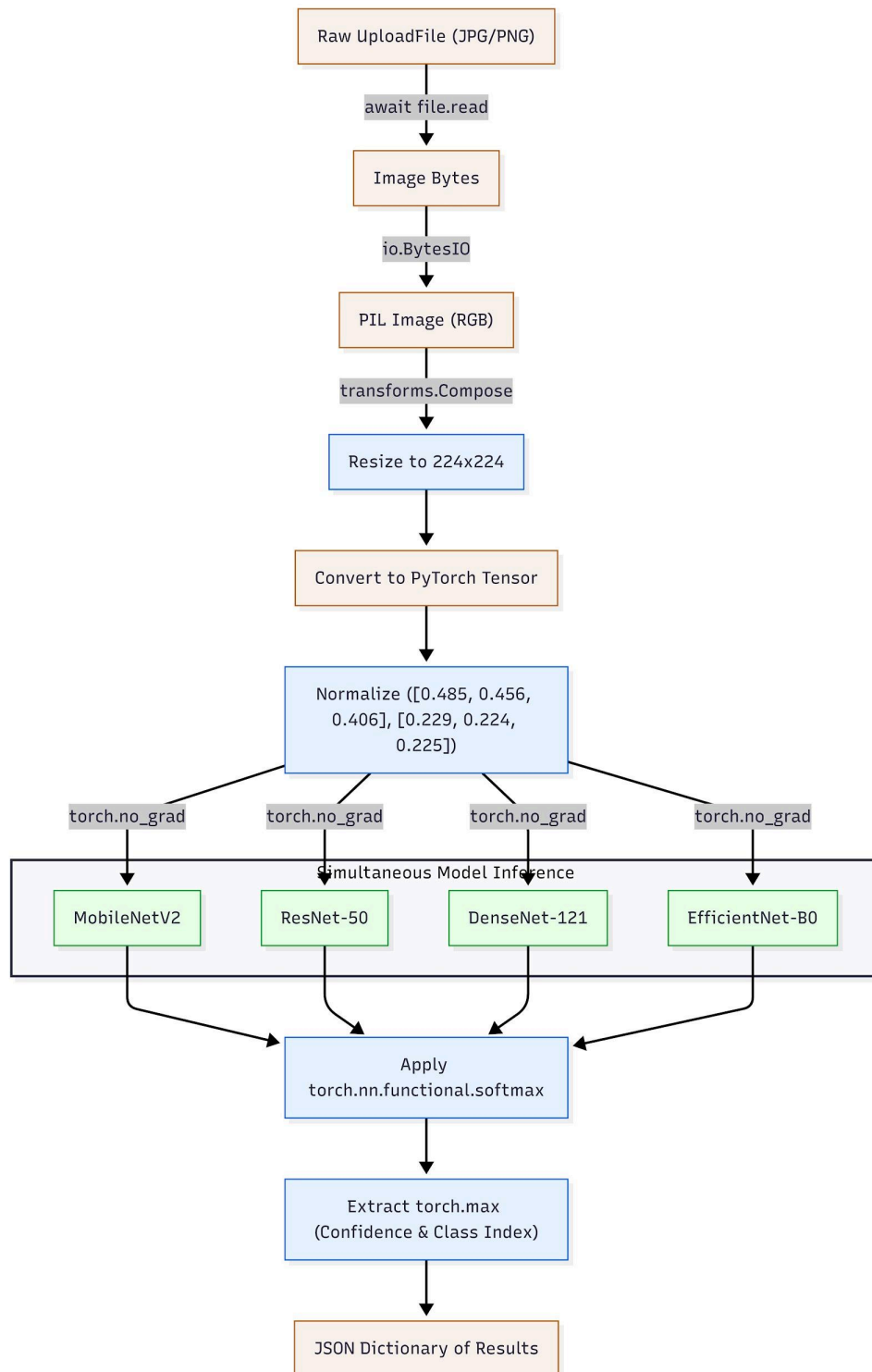


Figure 1.2: Machine Learning Data Pipeline (Data Transformation)

EXPERIMENTAL RESULTS AND MODEL EVALUATION

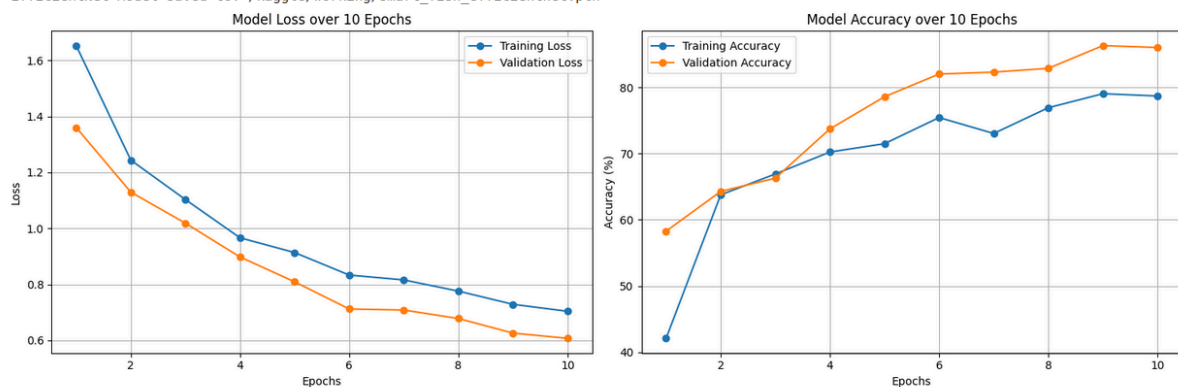
1. EfficientNet-B0

```

--- Initializing EfficientNet-B0 ---
Downloading: "https://download.pytorch.org/models/efficientnet_b0_rwightman-7f5810bc.pth" to /root/.cache/torch/hub/checkpoints/efficientnet_b0_rwightman-7f5810bc.pth
100%|██████████| 20.5M/20.5M [00:00<00:00, 139MB/s]
--- Starting EfficientNet-B0 Training ---
Epoch [1/10] Train Loss: 1.6530, Train Acc: 42.11% | Val Loss: 1.3606, Val Acc: 58.29%
Epoch [2/10] Train Loss: 1.2439, Train Acc: 63.77% | Val Loss: 1.1296, Val Acc: 64.29%
Epoch [3/10] Train Loss: 1.1031, Train Acc: 66.91% | Val Loss: 1.0187, Val Acc: 66.29%
Epoch [4/10] Train Loss: 0.9665, Train Acc: 70.23% | Val Loss: 0.8980, Val Acc: 73.71%
Epoch [5/10] Train Loss: 0.9137, Train Acc: 71.49% | Val Loss: 0.8092, Val Acc: 78.57%
Epoch [6/10] Train Loss: 0.8334, Train Acc: 75.43% | Val Loss: 0.7122, Val Acc: 82.00%
Epoch [7/10] Train Loss: 0.8159, Train Acc: 73.03% | Val Loss: 0.7084, Val Acc: 82.29%
Epoch [8/10] Train Loss: 0.7760, Train Acc: 76.91% | Val Loss: 0.6781, Val Acc: 82.86%
Epoch [9/10] Train Loss: 0.7287, Train Acc: 79.03% | Val Loss: 0.6260, Val Acc: 86.29%
Epoch [10/10] Train Loss: 0.7040, Train Acc: 78.69% | Val Loss: 0.6073, Val Acc: 86.00%
--- Training Complete! ---

```

EfficientNet Model saved to: /kaggle/working/smart_fish_efficientnet.pth



```

--- Starting EfficientNet Final Test Evaluation ---
EfficientNet Overall Test Accuracy: 82.00%

```

```

--- EfficientNet Accuracy per Disease Category ---

```

```

Bacterial Red disease      : 82.0% (41/50)
Bacterial diseases - Aeromoniasis : 86.0% (43/50)
Bacterial gill disease     : 82.0% (41/50)
Fungal diseases Saprolegniasis : 64.0% (32/50)
Healthy Fish               : 98.0% (49/50)
Parasitic diseases         : 98.0% (49/50)
Viral diseases White tail disease : 64.0% (32/50)

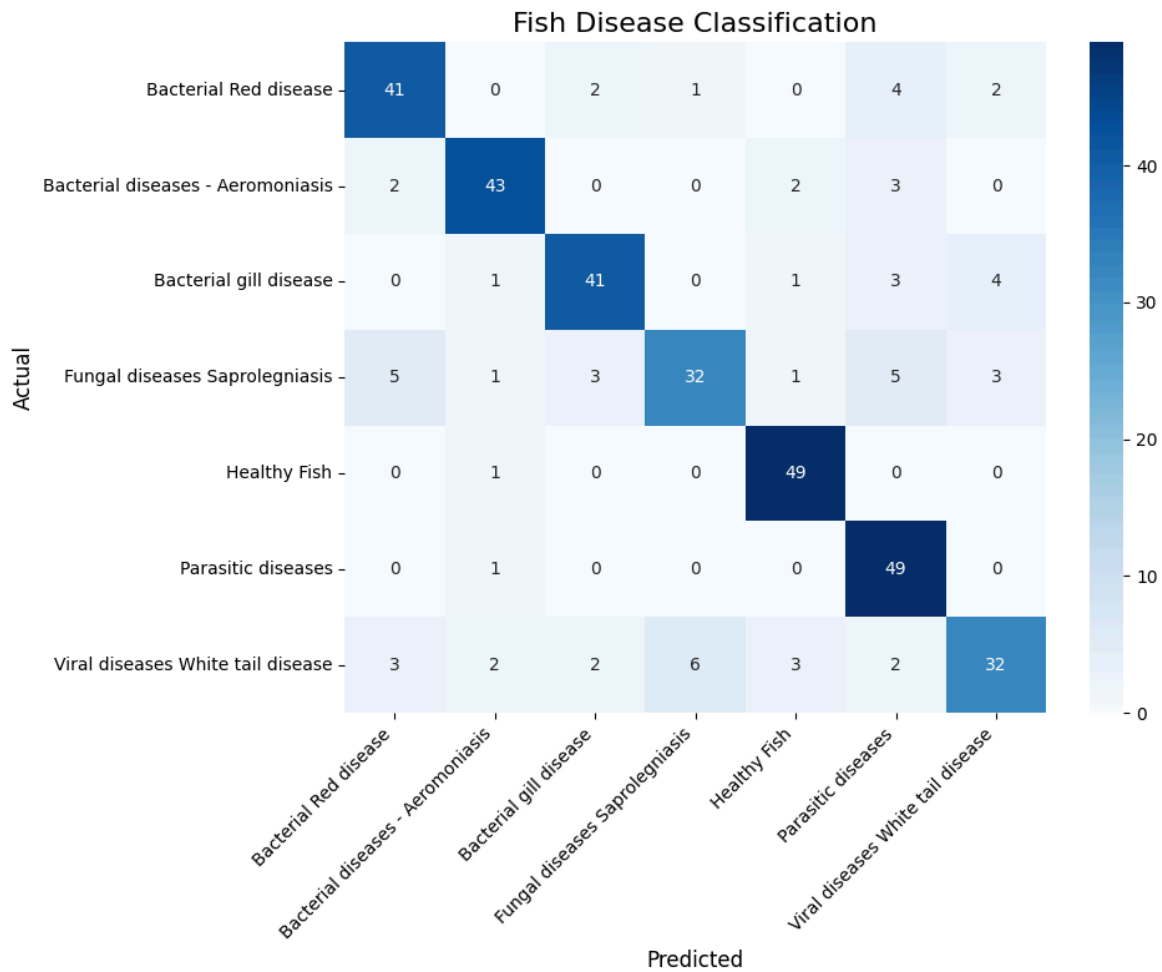
```

```

--- Detailed Classification Report ---

```

	precision	recall	f1-score	support
Bacterial Red disease	0.80	0.82	0.81	50
Bacterial diseases - Aeromoniasis	0.88	0.86	0.87	50
Bacterial gill disease	0.85	0.82	0.84	50
Fungal diseases Saprolegniasis	0.82	0.64	0.72	50
Healthy Fish	0.88	0.98	0.92	50
Parasitic diseases	0.74	0.98	0.84	50
Viral diseases White tail disease	0.78	0.64	0.70	50
accuracy			0.82	350
macro avg	0.82	0.82	0.82	350
weighted avg	0.82	0.82	0.82	350



2. DenseNet-121

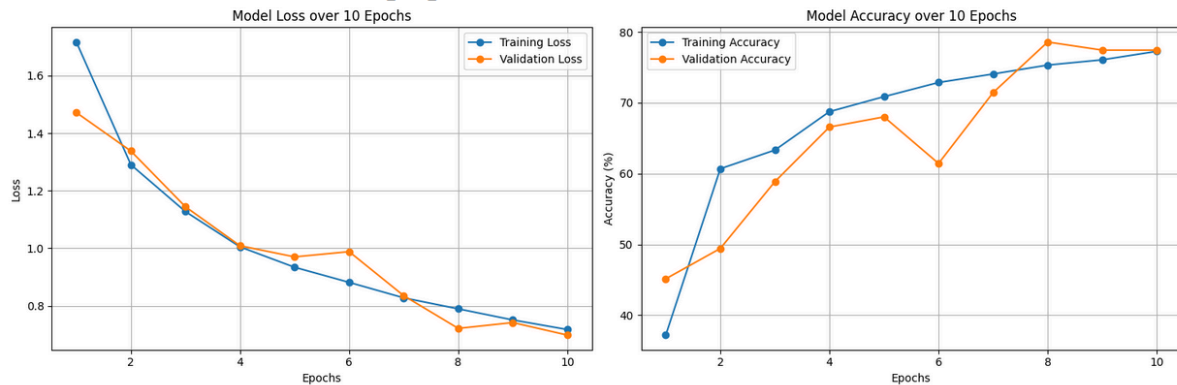
```

--- Initializing DenseNet-121 ---
Downloading: "https://download.pytorch.org/models/densenet121-a639ec97.pth" to /root/.cache/torch/hub/checkpoints/densenet121-a639ec97.pth
100%|██████████| 30.8M/30.8M [00:00<00:00, 205MB/s]
--- Starting DenseNet-121 Training ---
Epoch [1/10] Train Loss: 1.7168, Train Acc: 37.20% | Val Loss: 1.4716, Val Acc: 45.14%
Epoch [2/10] Train Loss: 1.2903, Train Acc: 60.69% | Val Loss: 1.3379, Val Acc: 49.43%
Epoch [3/10] Train Loss: 1.1275, Train Acc: 63.31% | Val Loss: 1.1442, Val Acc: 58.86%
Epoch [4/10] Train Loss: 1.0049, Train Acc: 68.74% | Val Loss: 1.0087, Val Acc: 66.57%
Epoch [5/10] Train Loss: 0.9341, Train Acc: 70.86% | Val Loss: 0.9701, Val Acc: 68.00%
Epoch [6/10] Train Loss: 0.8815, Train Acc: 72.86% | Val Loss: 0.9883, Val Acc: 61.43%
Epoch [7/10] Train Loss: 0.8278, Train Acc: 74.06% | Val Loss: 0.8353, Val Acc: 71.43%
Epoch [8/10] Train Loss: 0.7895, Train Acc: 75.31% | Val Loss: 0.7216, Val Acc: 78.57%
Epoch [9/10] Train Loss: 0.7510, Train Acc: 76.06% | Val Loss: 0.7419, Val Acc: 77.43%
Epoch [10/10] Train Loss: 0.7176, Train Acc: 77.26% | Val Loss: 0.6983, Val Acc: 77.43%

```

--- Training Complete! ---

DenseNet Model saved to: /kaggle/working/smart_fish_densenet.pth

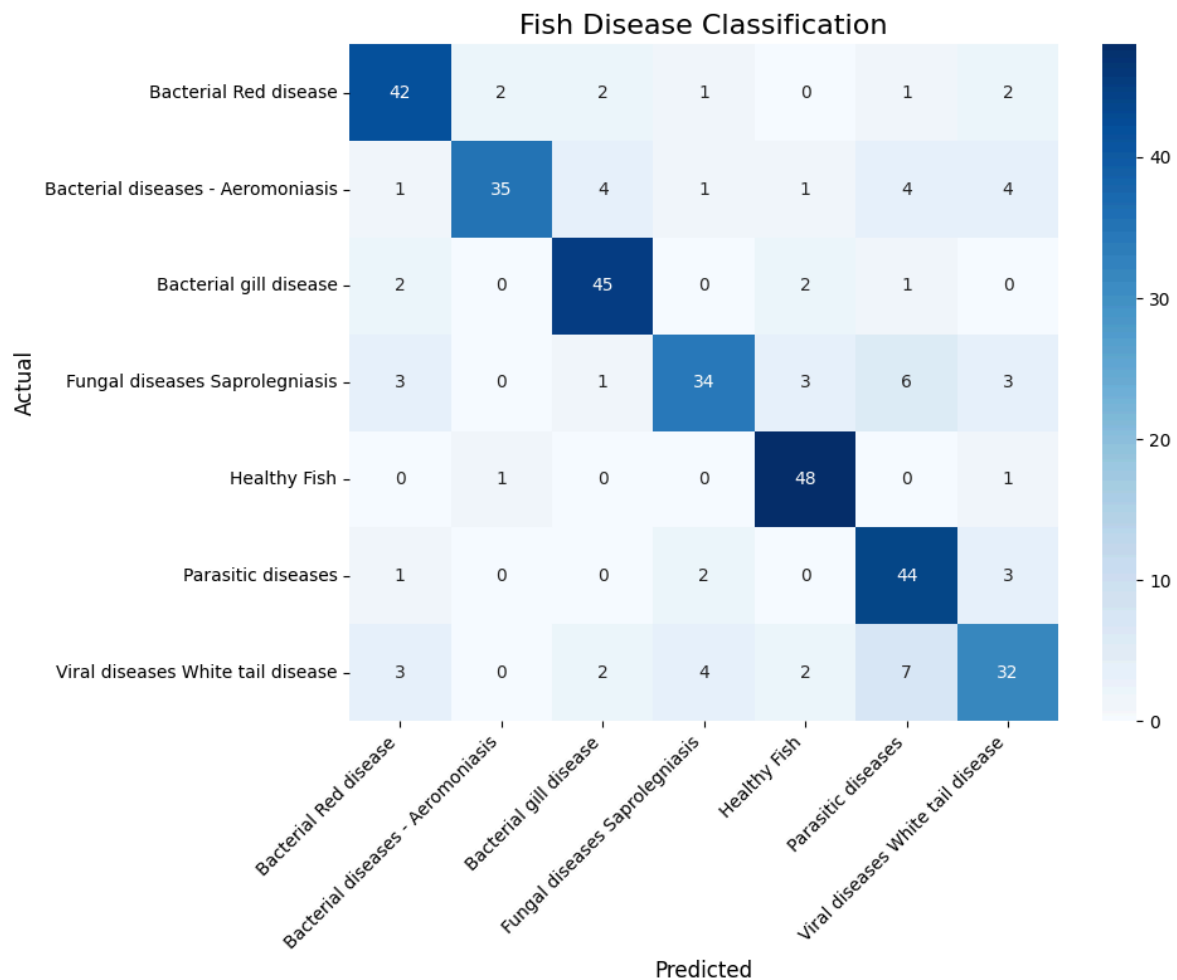


--- Starting DenseNet Final Test Evaluation ---
DenseNet Overall Test Accuracy: 80.00%

--- DenseNet Accuracy per Disease Category ---
Bacterial Red disease : 84.0% (42/50)
Bacterial diseases - Aeromoniasis : 70.0% (35/50)
Bacterial gill disease : 90.0% (45/50)
Fungal diseases Saprolegniasis : 68.0% (34/50)
Healthy Fish : 96.0% (48/50)
Parasitic diseases : 88.0% (44/50)
Viral diseases White tail disease : 64.0% (32/50)

--- Detailed Classification Report ---

	precision	recall	f1-score	support
Bacterial Red disease	0.81	0.84	0.82	50
Bacterial diseases - Aeromoniasis	0.92	0.70	0.80	50
Bacterial gill disease	0.83	0.90	0.87	50
Fungal diseases Saprolegniasis	0.81	0.68	0.74	50
Healthy Fish	0.86	0.96	0.91	50
Parasitic diseases	0.70	0.88	0.78	50
Viral diseases White tail disease	0.71	0.64	0.67	50
accuracy			0.80	350
macro avg	0.81	0.80	0.80	350
weighted avg	0.81	0.80	0.80	350



3. ResNet-50

```

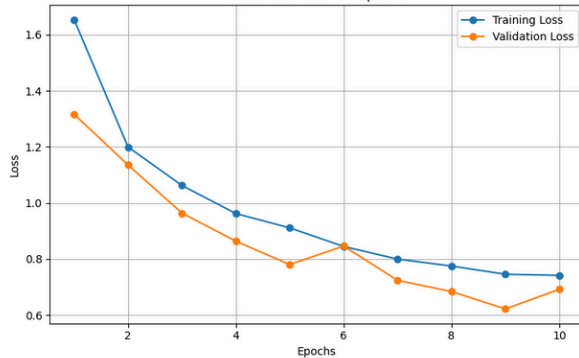
--- Initializing ResNet-50 ---
--- Starting ResNet-50 Training ---
Epoch [1/10] Train Loss: 1.6535, Train Acc: 38.40% | Val Loss: 1.3163, Val Acc: 53.14%
Epoch [2/10] Train Loss: 1.1995, Train Acc: 61.20% | Val Loss: 1.1356, Val Acc: 59.14%
Epoch [3/10] Train Loss: 1.0623, Train Acc: 64.51% | Val Loss: 0.9640, Val Acc: 65.71%
Epoch [4/10] Train Loss: 0.9623, Train Acc: 65.89% | Val Loss: 0.8638, Val Acc: 71.71%
Epoch [5/10] Train Loss: 0.9111, Train Acc: 69.77% | Val Loss: 0.7805, Val Acc: 74.86%
Epoch [6/10] Train Loss: 0.8449, Train Acc: 72.11% | Val Loss: 0.8463, Val Acc: 67.71%
Epoch [7/10] Train Loss: 0.7998, Train Acc: 74.11% | Val Loss: 0.7242, Val Acc: 76.29%
Epoch [8/10] Train Loss: 0.7750, Train Acc: 73.77% | Val Loss: 0.6842, Val Acc: 76.86%
Epoch [9/10] Train Loss: 0.7457, Train Acc: 74.06% | Val Loss: 0.6222, Val Acc: 80.00%
Epoch [10/10] Train Loss: 0.7418, Train Acc: 75.09% | Val Loss: 0.6920, Val Acc: 72.29%

--- Training Complete! ---

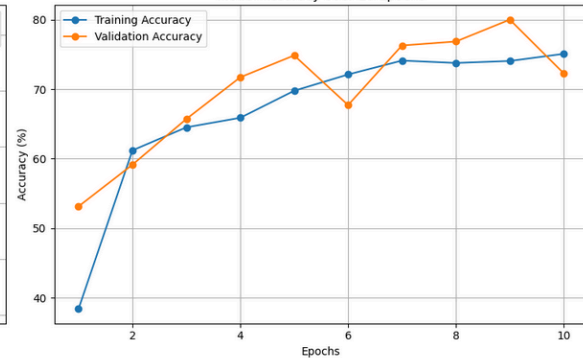
```

ResNet-50 Model saved to: /kaggle/working/smart_fish_resnet50.pth

Model Loss over 10 Epochs



Model Accuracy over 10 Epochs



```

--- Starting ResNet-50 Final Test Evaluation ---
ResNet-50 Overall Test Accuracy: 81.71%

```

```

--- ResNet-50 Accuracy per Disease Category ---

```

```

Bacterial Red disease      : 82.0% (41/50)
Bacterial diseases - Aeromoniasis : 90.0% (45/50)
Bacterial gill disease     : 90.0% (45/50)
Fungal diseases Saprolegniasis : 58.0% (29/50)
Healthy Fish               : 98.0% (49/50)
Parasitic diseases        : 72.0% (36/50)
Viral diseases White tail disease : 82.0% (41/50)

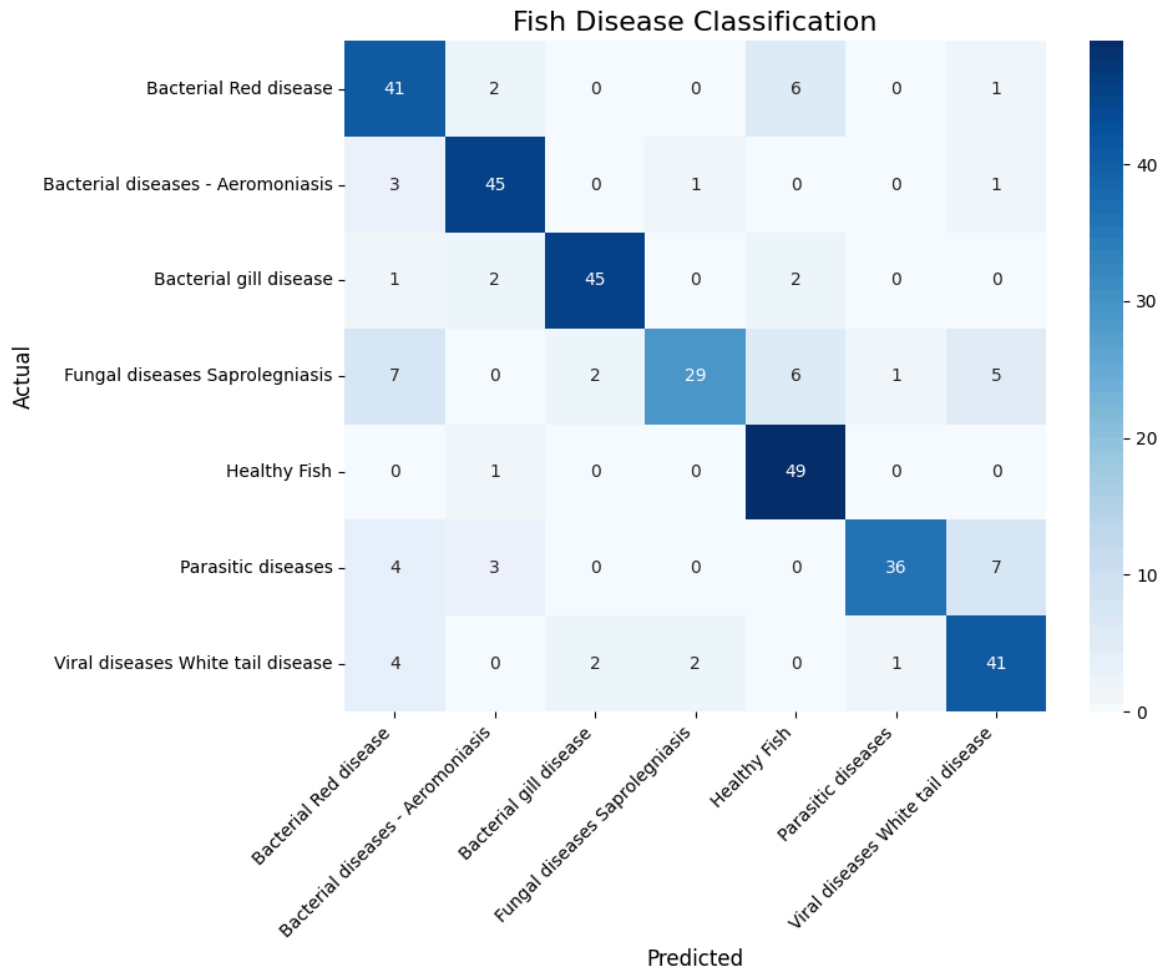
```

```

--- Detailed Classification Report ---

```

	precision	recall	f1-score	support
Bacterial Red disease	0.68	0.82	0.75	50
Bacterial diseases - Aeromoniasis	0.85	0.90	0.87	50
Bacterial gill disease	0.92	0.90	0.91	50
Fungal diseases Saprolegniasis	0.91	0.58	0.71	50
Healthy Fish	0.78	0.98	0.87	50
Parasitic diseases	0.95	0.72	0.82	50
Viral diseases White tail disease	0.75	0.82	0.78	50
accuracy			0.82	350
macro avg	0.83	0.82	0.81	350
weighted avg	0.83	0.82	0.81	350



4. MobileNetV2

```

--- Starting Training ---
Epoch [1/10] Train Loss: 1.6074, Train Acc: 40.57% | Val Loss: 1.2962, Val Acc: 52.00%
Epoch [2/10] Train Loss: 1.1860, Train Acc: 61.31% | Val Loss: 1.0373, Val Acc: 64.29%
Epoch [3/10] Train Loss: 1.0203, Train Acc: 67.71% | Val Loss: 0.8786, Val Acc: 70.57%
Epoch [4/10] Train Loss: 0.9622, Train Acc: 68.23% | Val Loss: 0.7456, Val Acc: 77.43%
Epoch [5/10] Train Loss: 0.8894, Train Acc: 71.37% | Val Loss: 0.7258, Val Acc: 76.57%
Epoch [6/10] Train Loss: 0.8449, Train Acc: 71.49% | Val Loss: 0.6786, Val Acc: 79.14%
Epoch [7/10] Train Loss: 0.8257, Train Acc: 72.11% | Val Loss: 0.6244, Val Acc: 80.57%
Epoch [8/10] Train Loss: 0.7679, Train Acc: 73.77% | Val Loss: 0.5892, Val Acc: 83.14%
Epoch [9/10] Train Loss: 0.7620, Train Acc: 73.77% | Val Loss: 0.5432, Val Acc: 84.29%
Epoch [10/10] Train Loss: 0.7353, Train Acc: 75.26% | Val Loss: 0.5640, Val Acc: 82.86%

```

--- Training Complete! ---

Model saved to: /kaggle/working/smart_fish_mobilenet.pth



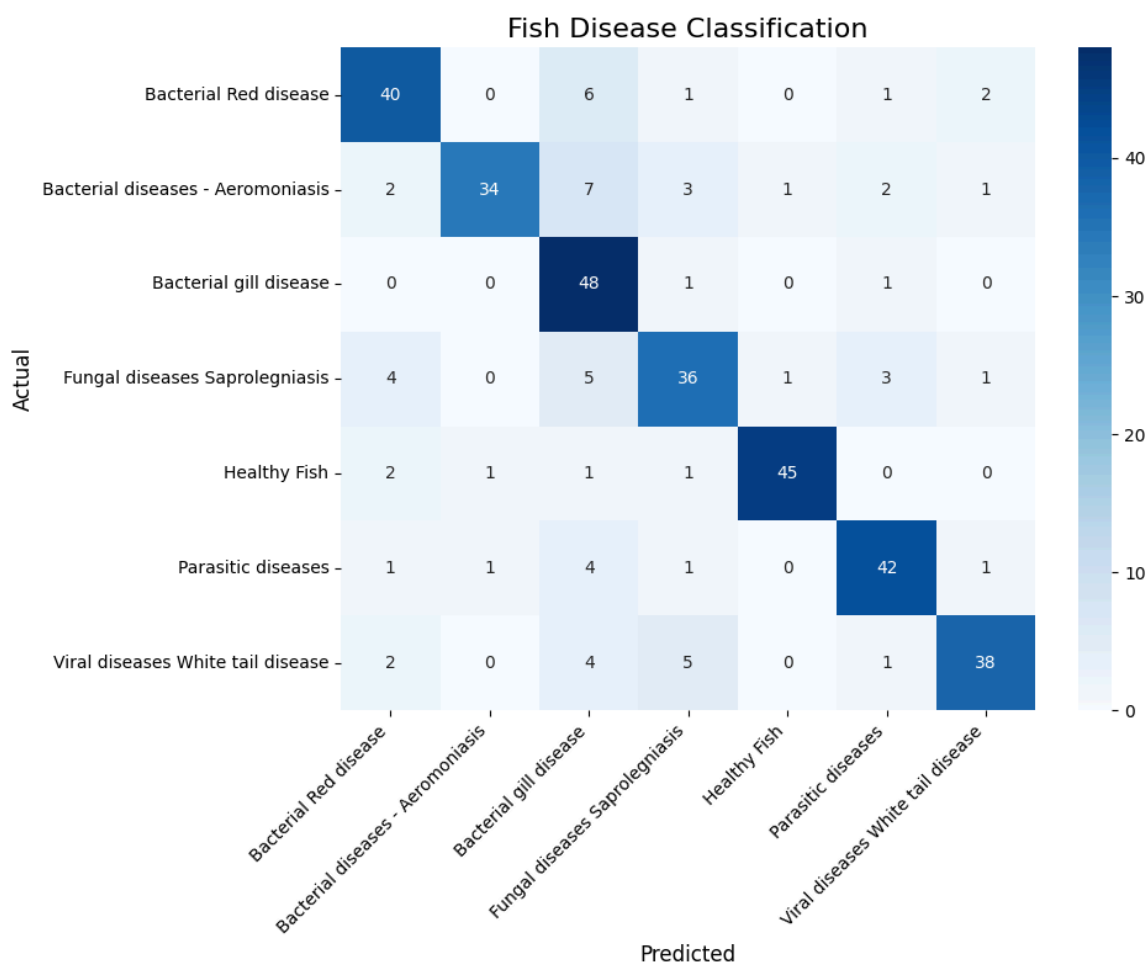
--- Starting Final Test Evaluation ---
Overall Test Accuracy: 80.86%

--- Accuracy per Disease Category ---

Bacterial Red disease : 80.0% (40/50)
Bacterial diseases - Aeromoniasis : 68.0% (34/50)
Bacterial gill disease : 96.0% (48/50)
Fungal diseases Saprolegniasis : 72.0% (36/50)
Healthy Fish : 90.0% (45/50)
Parasitic diseases : 84.0% (42/50)
Viral diseases White tail disease : 76.0% (38/50)

--- Detailed Classification Report ---

	precision	recall	f1-score	support
Bacterial Red disease	0.78	0.80	0.79	50
Bacterial diseases - Aeromoniasis	0.94	0.68	0.79	50
Bacterial gill disease	0.64	0.96	0.77	50
Fungal diseases Saprolegniasis	0.75	0.72	0.73	50
Healthy Fish	0.96	0.90	0.93	50
Parasitic diseases	0.84	0.84	0.84	50
Viral diseases White tail disease	0.88	0.76	0.82	50
accuracy			0.81	350
macro avg	0.83	0.81	0.81	350
weighted avg	0.83	0.81	0.81	350



5. Custom CNN

```

--- Starting Training ---
Epoch [1/10] Train Loss: 2.0433, Train Acc: 18.74% | Val Loss: 1.7962, Val Acc: 30.86%
Epoch [2/10] Train Loss: 1.7332, Train Acc: 33.14% | Val Loss: 1.5626, Val Acc: 39.71%
Epoch [3/10] Train Loss: 1.5889, Train Acc: 38.91% | Val Loss: 1.4110, Val Acc: 48.29%
Epoch [4/10] Train Loss: 1.4591, Train Acc: 44.57% | Val Loss: 1.3295, Val Acc: 52.86%
Epoch [5/10] Train Loss: 1.3811, Train Acc: 47.83% | Val Loss: 1.1403, Val Acc: 60.00%
Epoch [6/10] Train Loss: 1.2818, Train Acc: 53.14% | Val Loss: 0.9734, Val Acc: 64.29%
Epoch [7/10] Train Loss: 1.1935, Train Acc: 55.43% | Val Loss: 0.9086, Val Acc: 69.43%
Epoch [8/10] Train Loss: 1.1136, Train Acc: 59.03% | Val Loss: 0.7142, Val Acc: 77.14%
Epoch [9/10] Train Loss: 0.9852, Train Acc: 65.54% | Val Loss: 0.5903, Val Acc: 80.00%
Epoch [10/10] Train Loss: 0.8852, Train Acc: 68.51% | Val Loss: 0.6794, Val Acc: 73.71%

--- Training Complete! ---

```

Custom Model saved to: /kaggle/working/smart_fish_custom_cnn.pth



```

--- Starting Final Test Evaluation ---
Overall Test Accuracy: 74.29%

```

```

--- Accuracy per Disease Category ---
Bacterial Red disease       : 56.0% (28/50)
Bacterial diseases - Aeromoniasis : 96.0% (48/50)
Bacterial gill disease      : 80.0% (40/50)
Fungal diseases Saprolegniasis : 28.0% (14/50)
Healthy Fish                : 98.0% (49/50)
Parasitic diseases         : 84.0% (42/50)
Viral diseases White tail disease : 78.0% (39/50)

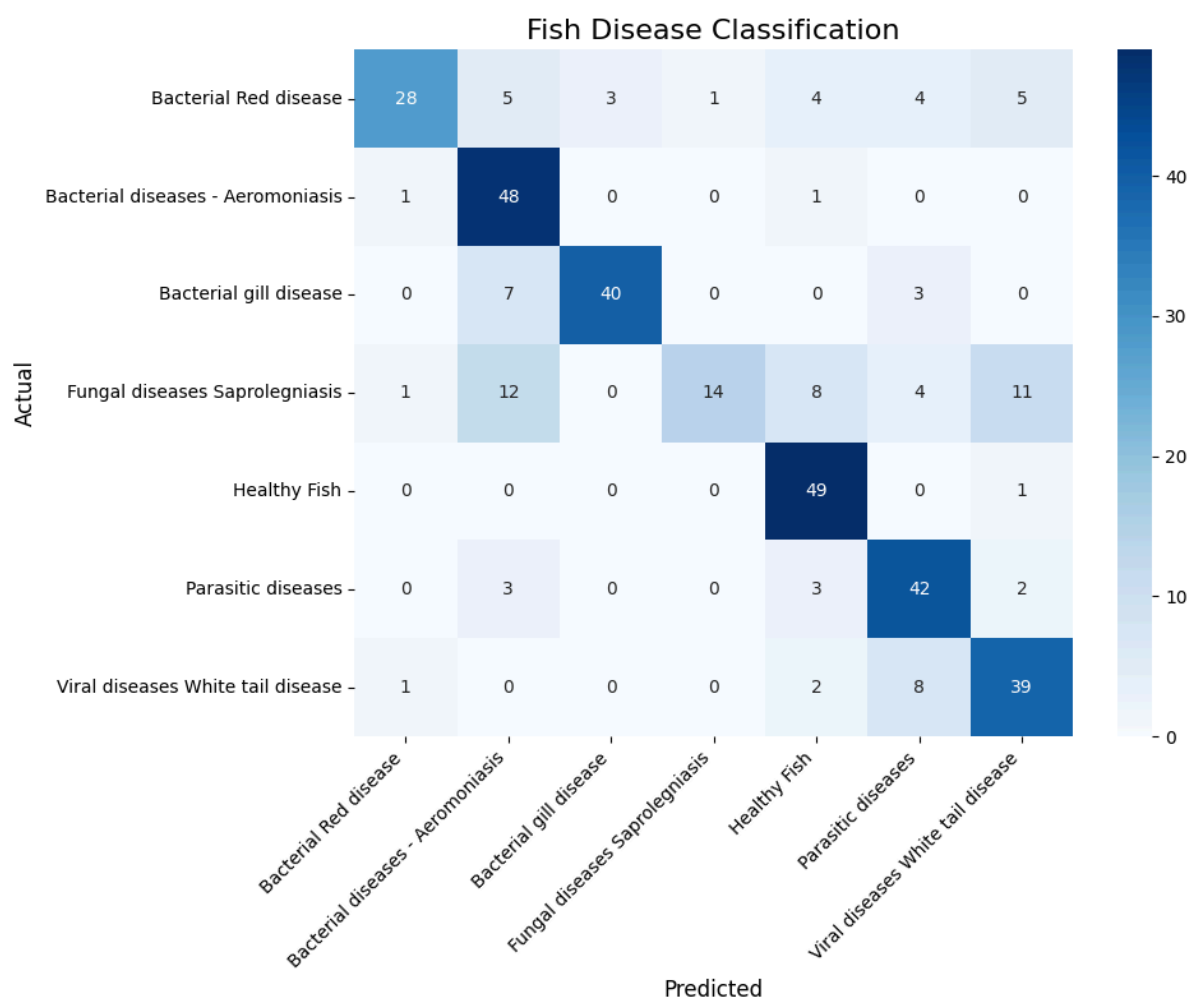
```

```

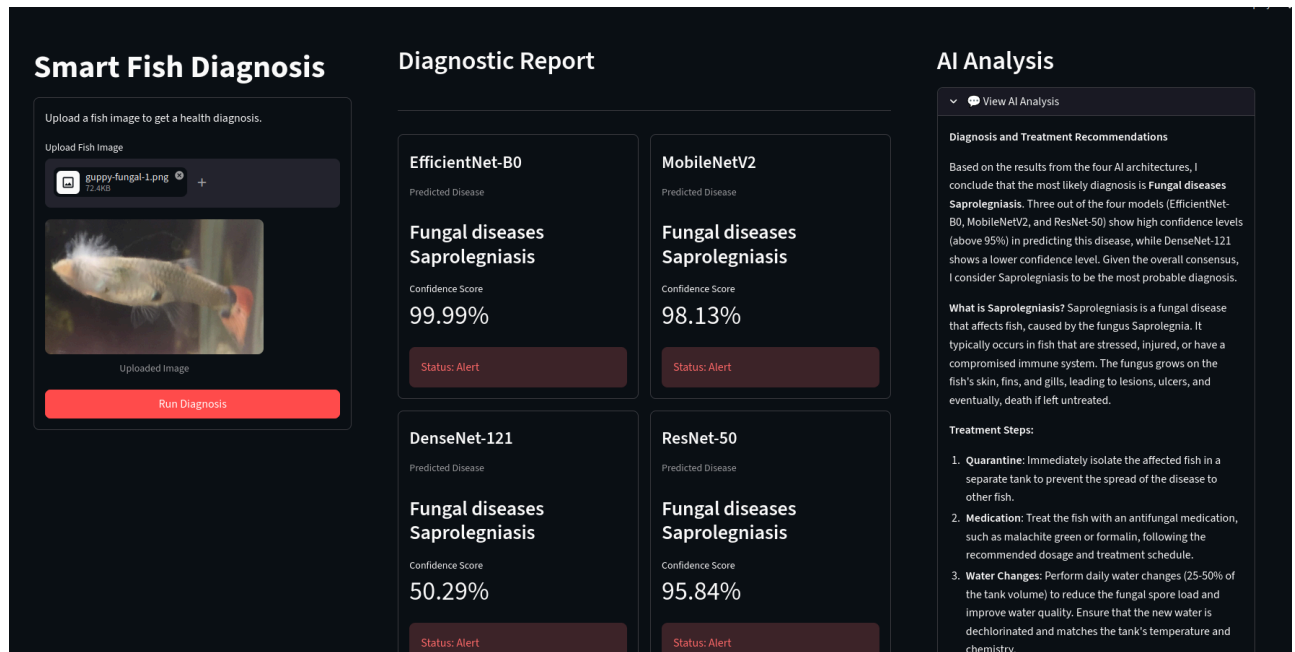
--- Detailed Classification Report ---

```

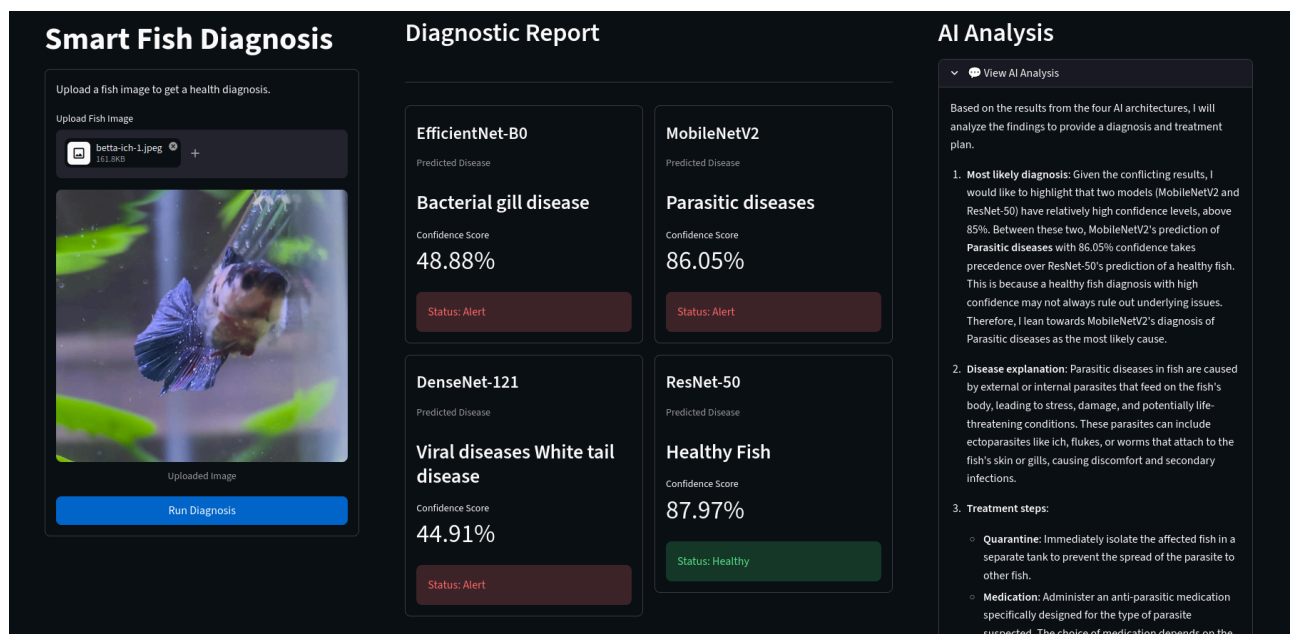
	precision	recall	f1-score	support
Bacterial Red disease	0.90	0.56	0.69	50
Bacterial diseases - Aeromoniasis	0.64	0.96	0.77	50
Bacterial gill disease	0.93	0.80	0.86	50
Fungal diseases Saprolegniasis	0.93	0.28	0.43	50
Healthy Fish	0.73	0.98	0.84	50
Parasitic diseases	0.69	0.84	0.76	50
Viral diseases White tail disease	0.67	0.78	0.72	50
accuracy			0.74	350
macro avg	0.79	0.74	0.72	350
weighted avg	0.79	0.74	0.72	350



APPLICATION INTERFACE



(Figure 2.1: A multi-model AI dashboard showing four CNN architectures (EfficientNet, MobileNetV2, DenseNet, and ResNet) correctly reaching a consensus diagnosis of fungal Saprolegniasis for an uploaded fish image)



(Figure 2.2: ResNet-50 was quite confident at 87.97% that the fish was healthy, the system was able to cross-reference the MobileNetV2 result to prioritize the parasitic diagnosis—likely a much safer approach for the fish!)

FUTURE SCOPE

1. IoT Integration and Real-Time Monitoring

- **Live Stream Diagnosis:** Moving beyond static image uploads by integrating live camera feeds (using Raspberry Pi or ESP32-CAM) directly into the aquarium or pond environment.
- **Sensor Fusion:** Combining visual data with water quality sensors (pH, temperature, dissolved oxygen, and ammonia levels). This would allow the system to predict disease outbreaks *before* physical symptoms appear, as poor water quality is the primary precursor to fish stress and infection.

2. Edge Deployment and Offline Capabilities

- **Mobile Optimization:** Utilizing **TensorFlow Lite** or **ONNX** to compress the models, allowing them to run locally on a mobile device. This is crucial for areas with limited or unstable internet connectivity.
- **Progressive Web App (PWA):** Converting the current dashboard into a PWA so it can be installed on a smartphone and used in the field without needing a browser.

REFERENCES

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